CONFIDENTIAL

UTM
UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING
UTM Johor Bahru

MIDTERM EXAMINATION

SUBJECT CODE	**: SECJ3323**
SUBJECT NAME	**: SOFTWARE ARCHITECTURE & DESIGN**
DATE / DAY	**: 17TH NOV 2025 / MONDAY**
START	**: 8.00 PM - 10.00 PM (2 HOURS)**
VENUE	**: BK1 - BK5, N24, UTM JB**

INSTRUCTION AND REMINDER

Answer all questions in the Answer Booklet. All diagrams must be drawn using the UML standard notations by hand. Note that any form of exam misconduct caught during the exam will cause students to get a zero in this assessment. This question booklet needs to be returned with the Answer Booklet and Answer Sheets(if any) by the end of the examination. You are only allowed to open this booklet with the permission of the examination invigilators' permission.

Fill in your particular below.

Name	
I/C No.	
Year / Course	
Section	
Lecturer Name	

THIS EXAMINATION PAPER CONSISTS OF 7 PRINTED PAGES INCLUDING THIS PAGE

SECTION A – MULTIPLE CHOICE QUESTIONS (20 Marks)

INSTRUCTIONS: ANSWER ALL THE FOLLOWING QUESTIONS IN ANSWER BOOKLET.

1. Which of the following best defines software design?
   A. The process of writing program code
   B. The process of transforming requirements into a blueprint for software construction.
   C. The process of testing a software system.
   D. The process of deploying a software product.

2. Why is modularity considered an important software design principle?
   A. It increases system size and complexity.
   B. It reduces dependencies and improves maintainability.
   C. It eliminates the need for documentation.
   D. It forces developers to use the same coding style.

3. Which UML diagram shows the structure of a system?
   A. Sequence Diagram
   B. Use Case Diagram
   C. Class Diagram
   D. Activity Diagram

4. How does a behavioral UML diagram differ from a structural UML diagram?
   A. Behavioral diagrams show system components; structural diagrams show interactions.
   B. Behavioral diagrams represent dynamic aspects; structural diagrams represent static aspects.
   C. Behavioral diagrams are optional; structural diagrams are mandatory.
   D. Behavioral diagrams focus on code structure only.

5. A system requires scalability and independent deployment of modules. Which architectural view should focus on deployment environments?
   A. Logical view
   B. Development view
   C. Process view
   D. Physical view

6. To improve the system design, an architect decides to add caching. This decision primarily addresses which quality attribute?
   A. Security
   B. Reliability
   C. Efficiency
   D. Portability

7. When analyzing a client-server architecture, which factor best explains its limitation?
   A. It offers high maintainability but low scalability if the server becomes a bottleneck.
   B. It offers low performance due to no separation of concerns.
   C. It prevents centralized control of data access.
   D. It increases client dependency on middleware.

8. Which criterion would you use to evaluate whether a microservices architecture is more suitable than a blackboard or repository architecture?
   A. The number of developers available.
   B. The frequency of system updates and deployment independence.
   C. The complexity of user interface design.
   D. The use of static configuration files.

9. A developer adds a new payment method to a system by extending an existing abstract Payment class instead of modifying the old code. Which principle is being applied?
   A. Liskov Substitution Principle
   B. Open-Closed Principle
   C. Single Responsibility Principle
   D. Dependency Inversion Principle

10. When analyzing a system design, you find that adding new report formats requires modifying existing classes instead of creating extensions. What does this indicate?
    A. The design violates the Open-Closed Principle because it requires code modification rather than extension.
    B. The design correctly applies the Liskov Substitution Principle.
    C. The design follows Open-Closed Principle but violates DRY (Don’t Repeat Yourself).
    D. The design demonstrates proper abstraction.

SECTION B – STRUCTURE QUESTIONS (48 MARKS)

INSTRUCTIONS: ANSWER ALL THE FOLLOWING QUESTIONS IN ANSWER BOOKLET.

Question 1 (16 Marks)

a) Describe any THREE (3) common design challenges faced during software design.

(6 Marks)

b) Explain the differences and contributions of a software architect and software engineers can affect the overall quality of a software design.

(4 Marks)

c) A development team is required to maintain a large system with high coupling and low cohesion. Explain the relationship of coupling and cohesion design principles and suggest one design strategy to improve the system structure and justify your choices.

(6 Marks)

Question 2 (16 Marks)

a) A Unified Modelling Language (UML) can be categorized into two types of diagrams. Name the TWO (2) category of diagrams and give a suitable example for each category.

(4 Marks)

b) Based on Figure 1, each relationship connector in the use case diagram has its own name and specification. Explain the specification of all the relationship connectors.

(4 Marks)

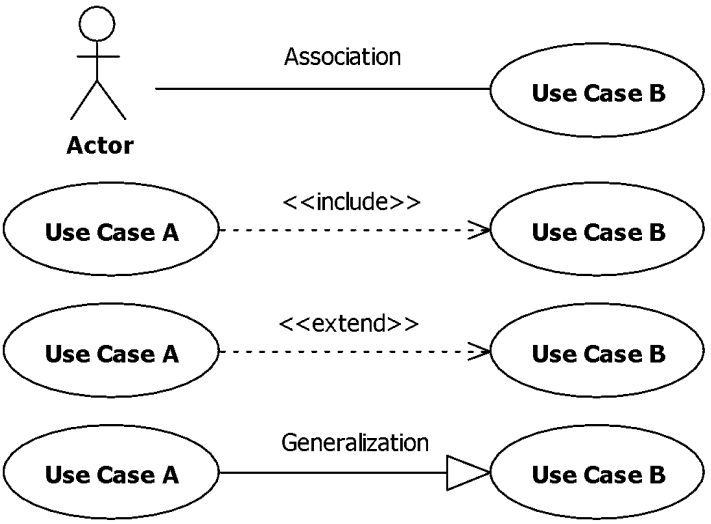


Figure 1: Sample Use Case Relationships

c) A class diagram can be used to automatically generate the code structure. Based on Figure 2, write the full code that can be generated in JAVA or C++ language.

(8 Marks)

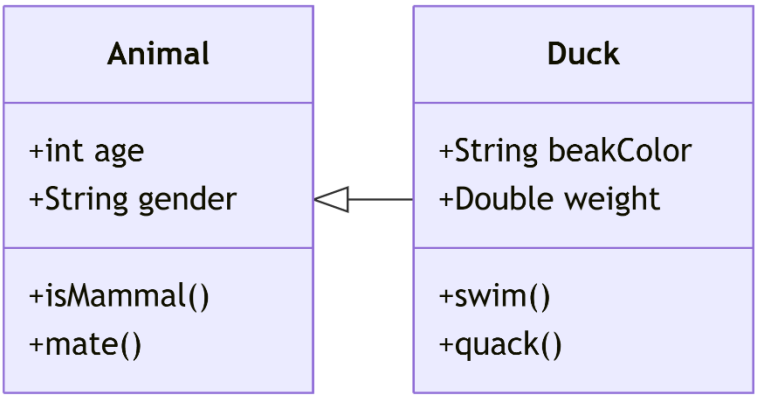


Figure 2: An Animal Class Diagram

Question 3 **(16 marks)**

a) Each software architecture is designed to have its own quality attribute. Identify and describe THREE (3) quality attributes in software design and architecture.

(3 Marks)

b) Identify and explain THREE (3) main tasks in the process of developing software architectures for a project.

(3 Marks)

c) Describe the application of the 4+1 architectural views model in formal software design documentation. For each view, specify its primary objective, intended stakeholders, and the underlying rationale for its distinct perspective on the system architecture?

(10 Marks)

SECTION B - PROBLEM SOLVING (32 Marks)

INSTRUCTIONS: ANSWER ALL THE FOLLOWING QUESTIONS IN ANSWER BOOKLET.

Project Description

A large online video platform is being developed to allow users to upload, store, and stream videos on demand. The system must support thousands of concurrent viewers, offer personalized content recommendations, and provide a smooth, adaptive streaming experience across multiple devices and network conditions. The platform should be modular so that independent teams can enhance various features such as content delivery, recommendation, and subscription management without interfering with core video playback functionality. To maintain flexibility, the system should allow new encoding algorithms or recommendation strategies to be added without modifying the core playback or recommendation engine code.

Quality Attributes

- Performance: High responsiveness during playback with efficient buffering and caching strategies.
- Scalability: Capable of handling growing numbers of users, video files, and concurrent streams.
- Availability: The streaming service should continue running even if some modules (e.g., analytics or recommendation) are temporarily unavailable.
- Modifiability: New video codecs or recommendation algorithms can be integrated easily.

Key Features

- Content Management Module for uploading, categorizing, and indexing video files.
- Streaming Engine that manages playback sessions, adaptive bitrate selection, and CDN integration.
- Recommendation Module that suggests videos based on watch history, ratings, and metadata.
- User Module for profile management, subscriptions, and watchlists.
- Analytics Module that tracks viewer statistics and playback performance.
- Admin Interface for moderation, content approval, and user management.

System Behaviors

- The Recommendation Engine and VideoEncoder modules are designed for extension — developers can add new algorithms (e.g., AI-based suggestions, new compression formats) without altering existing engine logic.
- Each video encoder subclass (e.g., H264Encoder, H265Encoder, AV1Encoder) can replace another while preserving the expected behavior and interface, ensuring compatibility in all encoding workflows.

a) Based on the project descriptions and requirements given, analyze and suggest a suitable architectural style for the project above.

(2 marks)

b) Evaluate the suggested architecture styles answered in (a), discuss how the architectural styles would address the requirements.

(4 marks)

c) Design and draw the class diagram to illustrate the suggested architecture style. The class diagram must reflect your architecture style answered in (a).

(8 marks)

d) Based on the system behaviors given, analyze and suggest a suitable detail design principles fit the behaviors.

(2 marks)

e) Evaluate and justify the suggested detail design principles answered in (d).

(4 marks)

f) Design and draw the class diagram to illustrate the suggested detail design principles. The class diagram must reflect your architecture style answered in (d).

(6 marks)

g) Evaluate and write the full code that can be generated from the class diagram drawn in (f).

(6 marks)

- QUESTIONS END -